# Yu-Sheng Shih

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## Current Position

**Research Assistant** — Sep. 2026–Present
Institute of Information Science, Academia Sinica — Taipei, Taiwan
*Supervisor:* Prof. Meng-Tsung Tsai

## Education

**M.S. in Computer Science and Information Engineering** — Jan. 2025–Aug. 2026
National Taiwan University — Taipei, Taiwan
*Advisor:* Prof. Shang-En Huang • Cumulative GPA: 4.08/4.3

**B.S. in Computer Science and Information Engineering** — Aug. 2019–Jun. 2024
National Taiwan University — Taipei, Taiwan
Double major in Mathematics • Cumulative GPA: 4.05/4.3

## Research Interests

Graph theory and graph algorithms; streaming and sublinear algorithms; communication complexity; extremal and probabilistic combinatorics; combinatorial optimization.

## Research Projects

**Space Complexity Dichotomies for Subgraph Finding Problems in the Streaming Model**
**Yu-Sheng Shih**, Meng-Tsung Tsai, Yen-Chu Tsai, and Ying-Sian Wu
Preprint • arXiv:2602.08002 (2026)
Established space-complexity dichotomies for subgraph and induced-subgraph finding in undirected and oriented graph streams.

**SDP-Based Reductions for Coloring 3-Colorable Graphs via Short Odd-Cycle Elimination**
Master's thesis • Completed Aug. 2026; related research ongoing
*Advisor:* Prof. Shang-En Huang
Developed SDP-based reductions that eliminate triangles and 5-cycles while retaining a large induced subgraph, reducing the coloring problem to instances of larger odd girth.

## Other Coauthored Work

Saintan Wu, Sen-Peng Eu, Kuo-Han Ku, and **Yu-Sheng Shih**. *Combinatorial proofs of Petrie Pieri rule and Plethystic Pieri rule.* *Annals of Combinatorics*, accepted for publication, 2026.

Tony Feng et al. (24 authors, including **Yu-Sheng Shih**). *Semi-Autonomous Mathematical Discovery with Gemini: A Case Study on the Erdős Problems.* arXiv:2601.22401, 2026.

## Awards

**International Mathematical Olympiad:** Silver Medal (2019); Bronze Medal (2018)

**ICPC Taipei Regional:** Silver Prize (2021 and 2022)

**Asian Pacific Mathematics Olympiad:** Bronze Medal (2018)

## Internships

**Research Intern, Streaming Algorithms** — Summer 2025
Institute of Information Science, Academia Sinica — Taipei, Taiwan

**Quantitative Research Intern** — Winter 2023
WorldQuant Taiwan — Taipei, Taiwan

## Teaching and Academic Service

**Teaching Assistant — Graph Theory with Applications** — Feb. 2026–Jun. 2026
National Taiwan University — Taipei, Taiwan

**Founder and Organizer, NTU TCS Study Group** — Sep. 2025–Present
National Taiwan University — Taipei, Taiwan
*Cofounders:* Peng-Ting Lin, Ben Cheung, Shang-En Huang, and Po-Yu Chou

**Teaching Assistant — Algorithm Design and Analysis** — Sep. 2021–Jan. 2022
National Taiwan University — Taipei, Taiwan

## Selected Academic Activities

**Selected Young Researcher** — Sep. 2026
13th Heidelberg Laureate Forum (HLF) — Heidelberg, Germany

**Participant, IMS Summer School** — Jul. 2026
National University of Singapore — Singapore
*Innovations and Challenges in Extremal Combinatorics*

**Taiwan Team Observer A, European Girls' Mathematical Olympiad** — Apr. 2026
EGMO — Bordeaux, France

**Participant, PCMI Graduate Summer School** — Jul. 2025
IAS/PCMI — Park City, Utah, USA
Program: *Extremal and Probabilistic Combinatorics*

## Languages

Mandarin Chinese (native); English (TOEFL iBT 101, 2024); Japanese (JLPT N2, 2026)